

SOYAL

Protocol

Model	F/W Version	Controller ID	Status Code
AR-401DI16	2.4	0x41	0x81
AR-401RO16	2.4	0x42	0x82
AR-401RO8DI16	2.4	0x44	0x85
AR-401DI4MO4	2.7	0x45	0x86

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Before connection:

You must confirm node ID and BAUD setting before connecting with AR-401ROxxDIxx.

	SW-8	SW-7	SW-6	
AR-401RO16	NO/NC (OFF/ON)	19200/9600 (ON/OFF)	4800/9600 (ON/OFF)	SW5~1:Node ID
AR-401RO8DI16	NO/NC (OFF/ON)	19200/9600 (ON/OFF)	SW 6~1:Node ID	
AR-401DI16				
AR-401DI4MO4			9600	

1. Communication Protocol

1.1 Preparing Connect

You must confirm node ID and BAUD setting before connecting with AR-401ROxxDIxx:

	SW-8	SW-7	SW-6	
AR-401RO16	NO/NC (OFF/ON)	19200/9600 (ON/OFF)	4800/9600 (ON/OFF)	SW5~1:Node ID
AR-401RO8DI16	NO/NC (OFF/ON)	19200/9600 (ON/OFF)	SW 6~1:Node ID	
AR-401DI16				

1.2 Data Format

All SOYAL® devices use the same protocol, only the commands given vary depending on different devices

Baud Rate			9600 , N , 8 , 1				
Data Format			Binary HEX Data				
Data Packet							
	Header	Length	Destination ID	Command	Data ...	XOR	SUM
Standard	8bits	8bits	8bits	8bits	0 ~244	8bits	8bits
Extend (RS485)	32 bits	16 bits	8bits	8bits	0~1018	8bits	8bits
Extend (TCP/IP)	32 bits	16 bits	8bits	8bits	0~1400	8bits	8bits

1.3 Data packed

1.3.1 Standard format

	Size (Byte)	Description	XOR 0xFF	SUM 0x00
Head	1	0x7E	No	No
Length	1	Data Length Indicator which denotes the length from Destination to the end including XOR and SUM (must less than 250 bytes)	No	No
Destination ID	1	Destination Node ID 00: Reserved for the bus master FF: Broadcast to each reader	Yes	Yes
Command Code	1	Instruction Command	Yes	Yes
Data	00 ~ F6	Length of Data Block which varies depending on instructions	Yes	Yes
XOR	1	To XOR each byte from Destination ID to Data with 0xFF		Yes
SUM	1	TO sum each byte from Destination to XOR with 0x00. If the summary is greater than 0xFF, it should keep the low byte.		

Example : Polling Status from Node. 1						
Head	Length	Destination	Command	Data ...	XOR	SUM
0x7E	0x04	0x01	0x18		0xE6	0xFF
		XOR Fields (Initial code:0xFF)				
		SUM Fields (Initial code:0x00)				

XOR = 0xFF ^ 0x01 ^ 0x18 = 0xE6

SUM = 0x01 + 0x18 + 0xE6 = 0xFF

Node ID needs to set from the

2. Command Code

2.1 Hosting Polling (18H) (Polling Controller Current Status)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	04	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination
Command	18	Hosting Polling
XOR	E6	XOR= FF^01^18 =E6
SUM	FF	SUM= 01+18+E6=FF

Echo:

	Value	
Head	7E	Leading Code
Length	??	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Destination	00	Destination Node ID. 00H mean send to host (PC or AR716E)
Ehco Type	xx	0x82=RO16
Source	01	Source Node ID. (Who send);
Data Field	Data 0	Bit7:6[DO3], Bit5:4[DO2], Bit3:2[DO1], Bit1:0[DO0]
	..	Bit7:6[DO7], Bit5:4[DO6], Bit3:2[DO5], Bit1:0[DO4]
		Bit7:6[DO11], Bit5:4[DO10], Bit3:2[DO9], Bit1:0[DO8]
		Bit7:6[DO15], Bit5:4[DO14], Bit3:2[DO13], Bit1:0[DO12]
XOR		Check byte
SUM		Check byte

DO Status: 0x00=Off, 0x01=On, 0x02: Flash at 1.67Hz

DI Status: 0x00=Open, 0x01=Close,(0x02:Break Line)

	Value	
Head	7E	Leading Code
Length	??	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Destination	00	Destination Node ID. 00H mean send to host (PC or AR716E)
Ehco Type	xx	0x85=RO8DI16
Source	01	Source Node ID. (Who send);
Data Field	Data 0	Bit7:6[DI3], Bit5:4[DI2], Bit3:2[DI1], Bit1:0[DI0],
	..	Bit7:6[DI7], Bit5:4[DI6], Bit3:2[DI5], Bit1:0[DI4]
		Bit7:6[DI11], Bit5:4[DI10], Bit3:2[DI9], Bit1:0[DI8]
	Data 3	Bit7:6[DI15], Bit5:4[DI14], Bit3:2[DI13], Bit1:0[DI12]
	Data 4	Bit7:Fire Input Status
	Data 5-20	16 bytes status change counter of each inputs.
	Data 21	Bit7:6[DO3], Bit5:4[DO2], Bit3:2[DO1], Bit1:0[DO0]
	Data 22	Bit7:6[DO7], Bit5:4[DO6], Bit3:2[DO5], Bit1:0[DO4]
	Data 23	Bit7:6[DO11], Bit5:4[DO10], Bit3:2[DO9], Bit1:0[DO8]
	Data 24	Bit7:6[DO15], Bit5:4[DO14], Bit3:2[DO13], Bit1:0[DO12]
XOR		Check byte
SUM		Check byte

2.2 Set Output Status (17H)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	xx	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination

Command	17	Set output status
Sub code		00h = Direct On/Off by Bitmap Table 01h = Latch On a specified output. 02h = Off a specified output. 03h = Flash a specified output at 1.67 Hz 04h = On a specified output with two bytes active interval on 10 time base. 05h = Flash On One Pp on 3Hz and limit total flash timer. 06h = Flash On One Pp on 3 Hz and limit pulse counter.
Parameters		Check the followed Parameters Table
...		
XOR		
SUM		

Parameters Table 1: Direct On/Off by Bitmap Table (00h):

P	Value	Description
Parameters	xx	Hi Byte of on Bit Table for output 15~8, Set Bit 0 to 1 will on output 8.
	xx	Lo byte of on Bit Table for output 7~0, Set Bit 0 to 1 will on output 0.
		Hi Byte of off Bit Table for output 15~8, Set Bit 0 to 1 will off output 8.
		Lo byte of off Bit Table for output 7~0, Set Bit 0 to 1 will off output 0.

Parameters Table 2: Latch On a specified output (01h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to on the specified output

Parameters Table 3: Off a specified output (02h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to on the specified output

Parameters Table 4: Flash a specified output at 1.67 Hz (03h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to flash a specified output

Parameters Table 5: On a specified output with two bytes active interval on 10 time base (04h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to active a specified interval
	xx	High byte of active interval
	xx	Low byte of active interval

Parameters Table 6: Flash a specified output with two bytes active interval on 10 time base (05h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to flash the specified output
	xx	High byte of active time interval
	xx	Low byte of active time interval

Parameters Table 7: Flash a specified output with two bytes active interval on 10 time base (06h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to flash the specified output
	xx	High byte of active pulse counter
	xx	Low byte of active pulse counter

Parameters Table 8: Set the specified output into PWM mode and base on 1ms (07h):

P	Value	Description
Parameters	xx	Range 00h ~ 0Fh to flash the specified output
	xx	High state duty time
	xx	Low state duty time

Echo:

	Value	
Head	7E	Leading Code
Length	??	Data Length Indicator which denotes the length from Node to the end including XOR and SUM
Destination	00	Destination Node ID. 00H mean send to host (PC or AR716E)
Ehco Type	xx	0x81=DI16,0x82=RO16,0x83=AI16,0x84=AO16,0x85=RO8DI16
Source	01	Source Node ID. (Who send);
Data Field	Data 0	Bit7:6[DO3], Bit5:4[DO2], Bit3:2[DO1], Bit1:0[DO0]
	..	Bit7:6[DO7], Bit5:4[DO6], Bit3:2[DO5], Bit1:0[DO4]
		Bit7:6[DO11], Bit5:4[DO10], Bit3:2[DO9], Bit1:0[DO8]
		Bit7:6[DO15], Bit5:4[DO14], Bit3:2[DO13], Bit1:0[DO12]
		Bit7:6[DI3], Bit5:4[DI2], Bit3:2[DI1], Bit1:0[DI0],
		Bit7:6[DI7], Bit5:4[DI6], Bit3:2[DI5], Bit1:0[DI4]
		Bit7:6[DI11], Bit5:4[DI10], Bit3:2[DI9], Bit1:0[DI8]
		Bit7:6[DI15], Bit5:4[DI14], Bit3:2[DI13], Bit1:0[DI12]
XOR		Check byte
SUM		Check byte

DO Status: 0x00=Off, 0x01=On, 0x02: Flash at 1.67Hz

DI Status: 0x00=Open, 0x01=Close, (0x02: Break Line)

2.3 Setup real time clock of target device (23H)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	0B	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination
Command	23	Setup real time clock
Data	00	Second
	01	Minute
	02	Hour
	03	Day of a week (SUN ~ SAT: 1~7)
	04	Day of month
	05	Month
	0B	Year % 100. (2011 = 0Bh)
XOR	DA	$XOR = FF^{*01^{*}23^{*}00^{*}01^{*}02^{*}03^{*}04^{*}05^{*}06} = DA$
SUM	13	$SUM = (01 + 23 + 00 + 01 + 02 + 03 + 04 + 05 + 06 + DA) \cdot FF = 13(LSB)$

Echo:

The controller will echo ACK or NACK to the source node.

2.4 Get real time clock, firmware version and controller mode code (24H)

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	0B	Data length can be one or two bytes depend on protocol format.
Node ID	01	Node ID of destination
Command	24	Get real time clock
XOR	DA	$XOR = FF \oplus 01 \oplus 24 = DA$
SUM	FF	$SUM = 01 + 24 + DA = FF$

Echo:

	Value	Description
Head	??	Leading Code could be standard format or extended format
Length	12	Data length can be one or two bytes depend on protocol format.
Node	00	The value 00 is fixed, the message would be sent to PC from the device
Function	03	Response to the request
Reader ID	01	Reader ID
Data Field	0A	Data 0 Second
	16	Data 1 Minute
	0D	Data 2 Hour
	06	Data 3 Day of a week (from 1 to 7 mapping to Sunday to Saturday)
	09	Data 4 Date
	0C	Data 5 Month
	05	Data 6 Year % 100. (2009 = 09)
	17	Data 7 Firmware Version (1V7)
	27	Data 8 Reserved.
	01	Data 9 Reserved.
	01	Data 10 Firmware Identify Code 00: Is standard firmware
	85	Data 11 Controller type ※Note:1
XOR	..	
SUM	..	

※Note1: Controller Type Code: R016=42h, DI16=43h, R8DI16=44h,